Volumes of Revolution

ANSWER KEY

The disk and washer methods

1 Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.
$V = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi.$

2 The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.
Bounds $x = -2$ to 2: $V = \pi\int_{-2}^{2} [4^2 - (x^2)^2]dx = \pi\int_{-2}^{2} (16 - x^4)\,dx = \pi\left[16x - \frac{x^5}{5}\right]_{-2}^{2} = \frac{384\pi}{5}.$

The shell method

3 The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.
$V = 2\pi\int_0^2 x \cdot x^2\,dx = 2\pi\int_0^2 x^3\,dx = 2\pi\left[\frac{x^4}{4}\right]_0^2 = 8\pi.$

4 The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.
$V = 2\pi\int_0^3 x \cdot x\,dx = 2\pi\left[\frac{x^3}{3}\right]_0^3 = 18\pi.$

Comparing methods

5 The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis (not the y-axis). Find the volume using the disk method.
$V = \pi\int_0^2 (x^2)^2\,dx = \pi\int_0^2 x^4\,dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}.$

6 The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x-axis. Find the volume using the washer method.
Intersections: $\sqrt{x} = x$ at $x = 0, 1$. On $[0, 1]$, $\sqrt{x} \geq x$, so the outer radius is $\sqrt{x}$, inner is x:
$V = \pi\int_0^1 [(\sqrt{x})^2 - x^2]dx = \pi\int_0^1 (x - x^2)\,dx = \pi\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{\pi}{6}.$

Volumes about other axes

7 The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.
$V = \pi\int_0^2 [(x^2 + 1)^2 - 1^2]dx = \pi\int_0^2 (x^4 + 2x^2)\,dx = \pi\left[\frac{x^5}{5} + \frac{2x^3}{3}\right]_0^2 = \pi\left(\frac{32}{5} + \frac{16}{3}\right) = \frac{176\pi}{15}.$

8 The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).
$V = 2\pi\int_0^2 (3 - x)x^2\,dx = 2\pi\int_0^2 (3x^2 - x^3)\,dx = 2\pi\left[x^3 - \frac{x^4}{4}\right]_0^2 = 2\pi(8 - 4) = 8\pi.$

Volumes with trigonometric and exponential curves

9 The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis. Find the volume, using $\sin^2 x = \frac{1-\cos 2x}{2}$.

$V = \pi\int_0^{\pi}\sin^2 x\,dx = \pi\int_0^{\pi}\frac{1-\cos 2x}{2}\,dx = \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi} = \frac{\pi}{2}(\pi - 0) = \frac{\pi^2}{2}$.

10 The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x-axis. Find the volume, to three decimal places.

$V = \pi\int_0^1 e^{-2x}\,dx = \pi\left[-\frac{1}{2}e^{-2x}\right]_0^1 = \frac{\pi}{2}(1 - e^{-2}) \approx \frac{\pi}{2}(0.8647) \approx 1.358$.

Setting up without evaluating

11 Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x-axis.

$V = \pi\int_1^e (\ln x)^2\,dx$.

12 Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y-axis, using the shell method.

$V = 2\pi\int_0^1 x(\sqrt{x} - x^2)dx$ (outer curve $\sqrt{x}$ minus inner curve x^2, since $\sqrt{x} \geq x^2$ on $[0, 1]$).

Visualizing the generating region

13 The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x-axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.

The solid is a paraboloid-like horn, narrow at $x = 0$ and widening to radius 2 at $x = 4$. A full cylinder of radius 2, length 4 would have volume $\pi(2)^2(4) = 16\pi$; since the paraboloid is narrower than the cylinder everywhere except at $x = 4$, a volume of 8π (exactly half the cylinder) is reasonable.

Volume of a solid with a hole (washer method practice)

14 The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x-axis. Find the volume using the washer method (outer radius x, inner radius x^3).

$V = \pi\int_0^1 [x^2 - x^6]dx = \pi\left[\frac{x^3}{3} - \frac{x^7}{7}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{7}\right) = \frac{4\pi}{21}$.

An applied volume problem

15 A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x-axis, where x is the height (cm) and y is the radius. Find the volume of the vase.

$V = \pi\int_0^4(\sqrt{x} + 1)^2\,dx = \pi\int_0^4(x + 2\sqrt{x} + 1)dx = \pi\left[\frac{x^2}{2} + \frac{4}{3}x^{3/2} + x\right]_0^4 = \pi\left(8 + \frac{32}{3} + 4\right) = \pi\left(\frac{68}{3}\right) = \frac{68\pi}{3}$ cm^3.

Volume with the shell method, revisited

16 The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y-axis. Use the shell method to find the volume.

$V = 2\pi\int_0^2 x(4 - x^2)\,dx = 2\pi\int_0^2(4x - x^3)\,dx = 2\pi\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2\pi(8 - 4) = 8\pi$.